

1 Informative dropout and trial-design choice in
2 aggregated N-of-1 biomarker-validation trials

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5 **Contents**

6	1 Abstract	2
7	2 Introduction	3
8	2.1 A claim about how N-of-1 trials should account for dropout	3
9	2.2 Why the question matters	5
10	2.3 What the published N-of-1 literature does and does not do	7
11	2.4 What this paper contributes	10
12	3 Informative-dropout model	12
13	4 The four design families and their vulnerability windows	15
14	5 Simulation design	18
15	6 Results	20
16	6.1 Power across designs and dropout patterns	21
17	6.2 Bias of the biomarker-treatment interaction estimate	22
18	6.3 Randomization-path decomposition	24
19	6.4 Sensitivity to dropout-pattern misspecification	24
20	7 Design-by-dropout recommendation	25
21	8 Discussion	27
22	8.1 Prototype Monte Carlo confirmation	27
23	9 Conclusions	30
24	10 Conflict of interest	33
25	11 Funding	33
26	12 Data availability statement	33
27	13 Reproducibility	33

29 1 Abstract

30 **Background.** Aggregated N-of-1 trial designs vary substantially in their robust-
 31 ness to informative dropout, yet the published methodological literature has doc-
 32 umented dropout as a generic problem without adequately engaging the coupling
 33 between the trial-design family (open-label (OL), open-label plus blinded discon-
 34 tinuation (OL+BDC), traditional crossover, hybrid) and the inferential conse-
 35 quences of informative dropout for the biomarker-treatment interaction estimand.
 36 The unaddressed coupling is not a minor sensitivity issue: the bias and power con-
 37 sequences of dropout differ by an order of magnitude across design families even
 38 at a fixed overall dropout rate, and the most powerful design under no dropout
 39 is not necessarily the most powerful design under dropout.

40 **Methods.** We formalize the informative-dropout model introduced in¹ (haz-
 41 ard depending on cumulative symptom worsening since baseline), place it within
 42 the classical missing-data taxonomy, and then reproduce and extend Hendrick-
 43 son’s four-design dropout-robustness analysis with wider parameter ranges, sen-
 44 sitivity to dropout-pattern misspecification at the analysis stage, and an explicit
 45 randomization-path decomposition. The simulation employs the existing pmsim-
 46 stats framework with extensions to the dropout-pattern grid, and reporting follows
 47 the² ADEMP framework.

48 **Results.** [Forthcoming.] Across all four design families, informative dropout
 49 that selectively retains treatment-non-responders appears to inflate the apparent
 50 biomarker-treatment interaction by underestimating the on-drug response of
 51 the responder stratum. The OL design is the most dropout-robust on power
 52 but contributes no useful biomarker-prediction information; the OL+BDC and
 53 hybrid N-of-1 designs are the most dropout-vulnerable on raw power but, para-
 54 doxically, recover power through the so-called ‘happy accident’ selection effect
 55 (informative dropout preferentially preserves the most-informative crossover
 56 blocks). The traditional crossover design is most robust on combined power-
 57 and-bias under symptom-worsening-driven dropout but most vulnerable under
 58 symptom-improvement-driven dropout, where treatment-responders drop early
 59 and the crossover signal is degraded.

60 **Conclusions.** Trial designers should evaluate proposed N-of-1 designs against
 61 the anticipated dropout-pattern mechanism, not merely against a generic dropout
 62 rate. We provide a design-by-dropout-pattern recommendation table that maps
 63 three classes of dropout mechanism (balanced, worsening-driven, improvement-
 64 driven) to preferred design choices.

2 Introduction

2.1 A claim about how N-of-1 trials should account for dropout

We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[1]

The paper advances two coupled claims about how N-of-1 design selection should account for informative dropout.

- The four design families (OL, OL+BDC, traditional crossover, hybrid) differ in dropout robustness in ways the published literature has not adequately characterized.
- Hendrickson's Figure 4A demonstrates power losses exceeding 50 percent in some design-by-dropout cells while remaining negligible in others.
- The dropout-by-design coupling is a first-order design choice, not a marginal sensitivity issue.

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We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[2]

The first claim is that design families differ in dropout robustness in ways the published literature has not adequately characterized.

- Power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells.
- Design rankings reverse depending on the dropout pattern.
- This is a first-order design choice, not a marginal sensitivity issue.

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The first is that the four candidate trial-design families (open-label, open-label plus blinded discontinuation, traditional crossover, and the hybrid N-of-1 design proposed by¹) differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterized. Hendrickson's Figure 4A heatmap demonstrates that the absolute power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells while remaining negligible in others, and the design rankings reverse depending on the dropout pattern. This is not a marginal sensitivity issue but a first-order design choice that the field has not, to the best of our

104 knowledge, foregrounded.

105 [3]

106 **The second claim is that the dropout mechanism, not merely**
107 **the rate, is the methodologically relevant object for design**
108 **selection.**

- 109 • Balanced, worsening-driven, and improvement-driven mecha-
110 nisms interact with design families in qualitatively different
111 ways.
- 112 • Trial designers should evaluate designs against the anticipated
113 dropout mechanism rather than a generic rate.
- 114 • The mechanism is the primary object; the rate is relatively super-
115 ficial.

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118 The second claim is that the dropout-pattern mechanism (balanced,
119 symptom-worsening-driven, symptom-improvement-driven) interacts
120 with the design family in qualitatively different ways than the overall
121 dropout rate alone, and that trial designers should therefore evalu-
122 ate proposed N-of-1 designs against the anticipated dropout-pattern
123 mechanism rather than against a generic dropout rate. That is, the
124 mechanism is the methodologically relevant object; the rate is rela-
125 tively superficial.

126 [4]

127 **The paper imports the mature missing-data taxonomy into**
128 **the N-of-1 design-comparative context for the first time.**

- 129 • The missing completely at random (MCAR), missing at random
130 (MAR), and missing not at random (MNAR) taxonomy and the
131 mixed-effects model for repeated measures (MMRM) have been
132 worked through for parallel-group RCTs but not for N-of-1 de-
133 signs comparatively.
- 134 • ICH E9 (R1) formalizes estimand-aware handling of intercurrent
135 events including dropout.
- 136 • The present paper undertakes the import of this machinery into
137 the N-of-1 design-comparative setting.

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140 The case is, in the broadest framing, a special application of the
141 missing-data taxonomy of Little and Rubin³ to the N-of-1 design class.
142 The taxonomy is mature: missing completely at random (MCAR),
143 missing at random (MAR), and missing not at random (MNAR) de-
144 fine a ladder of assumptions about the missingness mechanism, and

each layer imposes correspondingly stronger inferential conditions^{4,5}. The classical clinical-trial literature has worked through the implications for parallel-group randomized trials⁶⁻⁸ and has converged on the mixed-effects model for repeated measures (MMRM) as the default analytic strategy when dropout is plausibly MAR. The ICH E9 (R1) addendum⁹ formalizes the estimand-aware framework that handles intercurrent events including dropout¹⁰. Unfortunately, none of this machinery has been imported into N-of-1 methodology in a design-comparative way. The present paper undertakes that import.

2.2 Why the question matters

[5]

Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class.

- Informative dropout can selectively destroy within-participant contrasts central to the N-of-1 estimand.
- Patient self-selection is correlated with latent biomarker-response components that are partly signal, not nuisance.
- The four canonical designs differ in their phase-specific vulnerability windows.

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Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class. We shall take each in turn.

[6]

The first reason is that informative dropout selectively destroys within-participant contrasts, not merely sample size.

- Unlike parallel-group RCTs, N-of-1 dropout removes specific within-participant contrasts central to the estimand.
- A participant lost before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation.
- A 30-percent phase-selective dropout can degrade interaction power more than 50-percent uniform dropout.

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The first reason is that the N-of-1 estimand depends on within-participant contrasts that informative dropout can selectively destroy. In a parallel-group RCT, dropout reduces the effective sample size but does not, under MAR, systematically distort the per-arm mean response that the analysis estimates. In an aggregated N-of-1 trial, dropout removes specific within-participant contrasts: a participant who drops

out after the open-label phase but before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation that the design is constructed to deliver. The information loss is not merely in the sample size but in the combinatorial structure of the design. A 30-percent overall dropout rate that selectively removes participants from the post-discontinuation phase can degrade the biomarker-treatment interaction power more than a 50-percent dropout rate that drops uniformly across phases.

[7]

The second reason is that patient self-selection is correlated with latent components that are partly signal, not nuisance.

- High-BR patients tend to remain on drug; high-PB patients may drop at discontinuation; negative-TV patients may drop from worsening.
- Each mechanism produces a different dropout pattern that interacts differently with the design family.
- The worsening- and improvement-driven patterns operationalize the responder-versus-non-responder distinction the biomarker literature is validating.

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The second reason is that patient self-selection into and out of the trial is correlated with the latent BR-PB-TV components in ways that are not nuisance but partly signal. Patients with strong pharmacological response (high *BR*) are likely to remain on drug and therefore to remain in the trial; patients with strong placebo response (high *PB*) may drop out at the blinded-discontinuation phase when their symptoms return on the silently-switched placebo, mistaking the return for trial failure; patients with adverse natural-history trajectories (negative *TV*) may drop out from worsening regardless of treatment. Each of these mechanisms produces a different dropout pattern, and each interacts with the design family in a different way. The 'symptom-worsening-driven' and 'symptom-improvement-driven' patterns of¹ are operationalizations of the responder-versus-non-responder distinction that the predictive-biomarker literature is endeavoring to validate; the dropout pattern itself is therefore not incidental.

[8]

The third reason is that each design family has distinct phase-specific vulnerability windows whose trade-offs this paper analyzes.

- OL is most dropout-robust but cannot identify the BR-versus-PB

- 227 split required for the predictive-biomarker question.
- 228 • OL+BDC introduces a discontinuation-phase window where re-
- 229 sponders may drop from symptom return.
- 230 • The hybrid design has the largest vulnerability surface; the trade-
- 231 off between identifiability and dropout vulnerability is the central
- 232 design choice.

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235 The third reason is that the four canonical designs differ in their vul-

236 nerability windows. The open-label design has no discontinuation-

237 driven dropout and is therefore most robust to dropout overall, but

238 it cannot identify the *BR*-versus-*PB* split and is the wrong design

239 for the predictive-biomarker question. The OL+BDC design supplies

240 identifiability of the split but introduces a discontinuation-phase vul-

241 nerability window where treatment-responders may drop out from

242 symptom return. The traditional crossover design has two discontin-

243 uation windows (one per crossover) and is more exposed to symptom-

244 improvement-driven dropout. The hybrid N-of-1 design has the largest

245 number of identifying contrasts at risk and therefore the largest set

246 of vulnerability windows. The trade-off between identifiability and

247 dropout vulnerability is the central design choice this paper analyzes.

248 2.3 What the published N-of-1 literature does and does not do

249 [9]

250 **The methodological N-of-1 corpus addresses dropout gener-**
251 **ically but does not engage the design-by-dropout coupling**
252 **this paper centers on.**

- 253 • Lillie et al. and Vohra et al. CONSORT extension codify report-
- 254 ing standards but do not prescribe analytic methods for informa-
- 255 tive dropout in N-of-1 contexts.
- 256 • Systematic reviews document that dropout rates are reported
- 257 without specifying mechanism or analytic treatment.
- 258 • Dropout is one of the leading methodological weaknesses in N-of-1
- 259 reporting audits.

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262 The methodological N-of-1 corpus addresses dropout as a generic chal-

263 lenge but does not engage the design-by-dropout coupling this paper

264 centers on. We briefly survey what has been broached and what has

265 not.

266 [10]

Existing N-of-1 reporting standards codify dropout disclosure but do not prescribe analytic methods for informative dropout.

- Lillie et al. and the Vohra CONSORT extension set reporting standards without addressing informative-dropout analysis.
- Systematic reviews find that substantial proportions of N-of-1 studies report rates without specifying mechanism or analytic treatment.
- Dropout is one of the leading methodological weaknesses in N-of-1 reporting audits.

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Lillie et al.¹¹ and the CONSORT extension of Vohra et al.¹² codify reporting standards that include dropout reporting but do not prescribe analytic methods for handling informative dropout specifically in N-of-1 contexts. The systematic reviews of Stunnenberg et al.¹³ and Nate-san Batley and colleagues¹⁴ document that substantial proportions of published N-of-1 studies report dropout rates without specifying the dropout mechanism or its analytic treatment, and that dropout is one of the leading methodological weaknesses identified in N-of-1 reporting audits.

[11]

Hendrickson (2020) is the foundational reference; this paper extends it along three axes.

- The extension uses wider parameter ranges (carryover half-lives, effect sizes, dropout rates) than the original.
- It examines sensitivity to dropout-pattern misspecification at the analysis stage, which the original assumed away.
- It offers an explicit decomposition of the happy-accident randomization-path selection effect, scoped for path-stratified production testing.

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¹ introduce the first quantitative analysis of dropout- by-design coupling in the aggregated N-of-1 context, with a hazard- based dropout model whose intensity depends on cumulative symptom worsening since baseline. Their Figure 4A shows the four-design power matrix as a function of dropout pattern, and their Section 3.4 bias- quantification establishes that under symptom-worsening-driven dropout the OL+BDC design develops a substantial negative bias in the biomarker-treatment interaction coefficient ($p < 0.0001$). The present paper takes¹ as the foundational reference and extends it

along three axes. First, we use wider parameter ranges (longer carryover half-lives, higher biomarker effect sizes, larger dropout rates) than the original analysis. Second, we examine sensitivity to dropout-pattern misspecification at the analysis stage:¹ assumed the analyst knows the dropout mechanism, but in practice the analyst may model dropout as MCAR while the truth is MNAR, or may model it as symptom-worsening-driven when it is improvement-driven, and the present paper quantifies the inferential cost of each misspecification. Third, we offer an explicit decomposition of the ‘happy accident’ randomization-path selection effect.¹ noted that informative dropout preferentially preserves the most-informative crossover blocks in OL+BDC and hybrid N-of-1 designs (because patients with inconclusive open-label response are more likely to continue to the later blocks); in the present prototype we offer a marginal account of this effect, and a path-stratified test is scoped for the production rerun.

[12]

The paper’s contribution is to import classical missing-data, joint-modeling, and selection-bias machinery into the N-of-1 design-comparative simulation context.

- Little-Rubin taxonomy, selection models, and pattern-mixture models are adapted to the N-of-1 setting.
- Hazard-coupled-to-longitudinal-trajectory joint-modeling underwrites the Hendrickson dropout model and its extension.
- None of this machinery has previously appeared in the N-of-1 literature in a design-comparative simulation study.

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The classical missing-data literature^{3,4,15} supplies the methodological scaffolding (MAR/MNAR taxonomy, selection models, pattern-mixture models, multiple imputation) that we adapt to the N-of-1 setting. The informative-dropout and joint-modeling machinery^{16–18} supplies the hazard- coupled-to-longitudinal-trajectory formulation that underwrites the¹ dropout model and its extension here, and the longitudinal mixed-models foundations¹⁹ supply the analytic backbone for the $\beta_{bm:D}$ inference under MAR- plausible regimes. The clinical-trial dropout literature^{6–8} supplies the parallel-group analogue against which the design-comparative results below should be read, and the selection-bias literature²⁰ supplies the template for thinking about how response-based exclusion or informative dropout damages internal and external validity. None of this machinery has been imported into the N-of-1 methodology literature in a design-comparative simulation study, and that import is the present paper’s contribution.

2.4 What this paper contributes

[13]

The paper makes four contributions across Sections 2 through 6.

- Section 2 formalizes the informative-dropout model and places it in the missing-data taxonomy.
- Section 3 characterizes each design family’s phase-specific vulnerability windows.
- Sections 4 and 5 conduct and report a Monte Carlo simulation crossing four designs against five dropout patterns.

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We make four contributions, each addressed in a section below.

[14]

Section 2 formalizes the hazard-based informative-dropout model and establishes its MNAR character explicitly.

- The Hendrickson hazard model is placed in the MCAR/MAR/MNAR taxonomy and shown to be MNAR.
- Four canonical dropout patterns plus a no-dropout reference are defined as cells in the (β_0, β_1) hazard-parameter space.
- The operational definitions fix the simulation’s data-generating mechanism for the subsequent studies.

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In Section 2 we formalize the informative-dropout model for aggregated N-of-1 trials. The hazard-based model of¹ is placed in the missing-data taxonomy, its MNAR character is established explicitly, and the four canonical dropout patterns (balanced, more-of-flat, more-of-biased, high-dropout, plus a fifth no-dropout reference) are defined operationally as parameter cells in the (β_0, β_1) space of the hazard.

[15]

Section 3 identifies the phase-specific dropout-vulnerability windows and at-risk within-participant contrasts for each design family.

- Each of the four design families is characterized by its exposure to each dropout pattern per phase.
- Within-participant contrasts that are at risk of removal under each pattern are identified explicitly.

- 390 • The characterization motivates the design-by-dropout-pattern
391 structure of the simulation in Sections 4 and 5.

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394 In Section 3 we characterize the four design families’ dropout vulner-
395 ability windows. For each of the open-label, OL+BDC, traditional
396 crossover, and hybrid N-of-1 designs we identify the phase-specific ex-
397 posure to each dropout pattern and the within-participant contrasts
398 that are at risk of being removed.

399 [16]

400 **Sections 4 and 5 report a Monte Carlo study crossing four**
401 **designs against five dropout patterns at three sample sizes**
402 **and three effect sizes.**

- 403 • Performance measures include power, Type I error, bias, empiri-
404 cal SE, and path-conditional power decomposition.
- 405 • Reporting follows the Morris (2019) ADEMP framework with
406 Monte Carlo standard errors (MCSE) alongside every estimate.
- 407 • The randomization-path decomposition quantifies the happy-
408 accident selection effect.

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411 In Sections 4 and 5 we conduct and report a Monte Carlo simulation
412 study that crosses the four design families against the five dropout
413 patterns at three total sample sizes ($N \in \{35, 70, 100\}$, allocated
414 across each design’s randomization paths) and three biomarker-effect
415 sizes ($\beta_{bm} \in \{0, 0.3, 0.6\}$). The simulation reports power, Type I er-
416 ror, bias of $\hat{\beta}_{bm:D}$, mean standard error, and the randomization-path-
417 conditional power decomposition that quantifies the ‘happy accident’
418 selection effect. Reporting follows the² ADEMP framework adopted
419 as the project-wide simulation-reporting standard.

420 [17]

421 **Section 6 synthesises a recommendation table mapping**
422 **dropout mechanism class to preferred N-of-1 design choice.**

- 423 • Three mechanism classes (balanced, worsening-driven,
424 improvement-driven) are mapped to preferred designs indexed
425 by anticipated dropout rate.
- 426 • The table is presented as practical guidance for trial designers
427 selecting among the four canonical N-of-1 families.
- 428 • Section 7 discusses implications for trial-design selection and pm-
429 simstats operating characteristics.

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432 In Section 6 we synthesise a design-by-dropout-pattern recommenda-
433 tion table that maps three classes of dropout mechanism (balanced,
434 worsening- driven, improvement-driven) to preferred design choices in-
435 dexed by the anticipated overall dropout rate. The recommendation
436 is presented as guidance for trial designers selecting among the four
437 canonical N-of-1 design families when a particular dropout mechanism
438 is anticipated.

439 [18]

440 **Section 7 discusses implications for trial-design selection and**
441 **pmsimstats operating characteristics.**

- 442 • The discussion addresses design selection for biomarker-
443 validation N-of-1 trials.
- 444 • Operating characteristics of the pmsimstats simulation program
445 are reviewed.

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448 Section 7 discusses the implications for trial-design selection in
449 biomarker-validation N-of-1 trials and for the operating characteris-
450 tics of the pmsimstats simulation program.

451 3 Informative-dropout model

452 [19]

453 **The censoring hazard combines a time-only ignorable com-**
454 **ponent with a symptom-trajectory-biased component, made**
455 **absorbing by a monotone carry-forward pass.**

- 456 • The per-timepoint censoring probability is the union of two inde-
457 pendent draws: a flat component scaling with elapsed time and a
458 biased component scaling with the convex-transformed symptom
459 change since baseline.
- 460 • Two hazard parameters govern the split: β_0 sets the overall
461 dropout fraction and β_1 the proportion of that fraction routed
462 through the symptom-biased component, with the convexity ex-
463 ponent e_{b1} fixed at two.
- 464 • A second carry-forward pass renders dropout absorbing: once a
465 participant is censored at a timepoint, all later timepoints for
466 that participant are censored.

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We adopt the hazard-based informative-dropout model implemented in the pmsimstats R/censordata.R infrastructure, which operationalizes the mechanism of¹. For participant i at timepoint t , the marginal censoring probability is the union (logical OR) of two independent Bernoulli draws,

$$\Pr(\text{drop}_{it}) = 1 - (1 - p_{it}^{\text{flat}})(1 - p_{it}^{\text{bias}}),$$

where the flat component $p_{it}^{\text{flat}} \propto t$ depends on elapsed time alone and is calibrated to a marginal dropout fraction of $\beta_0(1 - \beta_1)$, and the biased component $p_{it}^{\text{bias}} \propto t(\Delta_{Sx}(t) + 100)^{e_{b1}}$ depends on the symptom change $\Delta_{Sx}(t)$ since baseline and is calibrated to $\beta_0\beta_1$. The additive shift of 100 keeps the convex argument positive across the realised range of symptom scores, so that a large improvement lowers the dropout probability while a small change or a worsening raises it. A second pass enforces monotone dropout: once a participant is censored at a timepoint, every later timepoint for that participant is also set missing. The hazard is thus governed by three quantities: the overall rate β_0 , the biased proportion β_1 , and the convexity exponent e_{b1} , held at two throughout.

[20]

Whenever the biased component is active the mechanism is missing-not-at-random, because the censoring indicator at a timepoint depends on the symptom value that the same indicator renders missing.

- The flat component depends only on elapsed time, a fully observed design variable, and is therefore ignorable with respect to the latent response.
- The biased component makes dropout at time t depend on $\Delta_{Sx}(t)$, and because that observation is itself censored at the triggering timepoint the mechanism is MNAR in the Little-Rubin sense.
- The departure from ignorability is controlled by β_1 : at $\beta_1 = 0$ the mechanism reduces to the ignorable limit, and the MNAR departure increases monotonically with β_1 .

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The two components occupy different rungs of the Little-Rubin ladder. The flat component depends only on elapsed time, which is a fully observed design variable rather than a function of the response, so on its own it induces missingness that is ignorable for likelihood-based inference. The biased component is the consequential one. Because the censoring indicator for the observation at time t is a function of $\Delta_{Sx}(t)$, the symptom change realised at that same timepoint, and because the

triggering observation is itself rendered missing in the carry-forward pass, the missingness of Y_{it} depends on the value of Y_{it} . This is the defining condition for missing-not-at-random data^{3,4}. The mechanism is therefore MNAR for every pattern with $\beta_1 > 0$, and the standard MMRM appeal to a MAR justification⁶ does not hold. The extent of the violation scales with β_1 , which makes the biased proportion, not the overall rate, the methodologically load-bearing parameter.

[21]

Five dropout patterns are fixed as cells in the (β_0, β_1) hazard-parameter space, spanning a no-dropout reference through balanced, flat-weighted, bias-weighted, and high-rate conditions.

- The no-dropout reference sets $\beta_0 = 0$ and supplies the full-data gold standard against which the bias estimands of Study 2 are measured.
- Balanced ($\beta_0 = 0.05, \beta_1 = 0.5$), more-of-flat ($\beta_1 = 0.2$), and more-of-biased ($\beta_1 = 0.8$) hold the overall rate fixed while varying informativeness from mostly-ignorable to strongly MNAR.
- High-dropout ($\beta_0 = 0.15, \beta_1 = 0.5$) raises the overall rate at the balanced split, isolating the rate axis from the informativeness axis.

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The five dropout patterns are defined operationally as cells in the (β_0, β_1) parameter space of the hazard, with $e_{b1} = 2$ held throughout. The no-dropout reference ($\beta_0 = 0$) retains the complete data and furnishes the gold-standard estimate of $\beta_{bm:D}$ used in the second bias-reference mode of Study 2. The balanced pattern ($\beta_0 = 0.05, \beta_1 = 0.5$) routes half of a modest five-percent dropout fraction through the symptom-biased component. Holding β_0 fixed, the more-of-flat ($\beta_1 = 0.2$) and more-of-biased ($\beta_1 = 0.8$) patterns move the same overall rate toward the ignorable and the strongly MNAR ends of the informativeness axis respectively, so that any difference in operating characteristics between them isolates the effect of the mechanism from the effect of the rate. The high-dropout pattern ($\beta_0 = 0.15, \beta_1 = 0.5$) triples the overall rate at the balanced split, isolating the complementary rate axis. The sign convention of the biased component is worsening-driven, in that a worsening relative to baseline raises the dropout probability; the symptom-improvement-driven variant examined in the design-recommendation of Section 6 is obtained by reflecting the symptom-change argument, so that improvement rather than worsening raises the hazard.

4 The four design families and their vulnerability windows

[22]

The four design families are distinguished by their phase structure, which determines both the within-participant contrasts that identify the biomarker-treatment interaction and the windows in which informative dropout can remove them.

- Each design is a set of randomization paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors of `buildtrialdesign`.
- The estimand-defining contrasts are within-participant differences across phases; informative dropout is damaging in proportion to how directly it removes those contrasts rather than merely reducing sample size.
- The four designs are ordered by an increasing number of identifying contrasts at risk, and therefore by an increasing number of vulnerability windows, from open-label through the hybrid N-of-1 design.

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Each candidate design is represented as a set of randomization paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors that `buildtrialdesign` accepts. The biomarker-treatment interaction is identified by within-participant contrasts across those phases, so the consequence of informative dropout for a given design is determined by which phase a dropout removes and which contrast that phase carries. We take the four designs in order of increasing number of identifying contrasts at risk.

[23]

The open-label design has a single all-on-drug path, no discontinuation window, and is the most dropout-robust but cannot identify the pharmacological-versus-placebo split.

- The single path carries only an on-drug-versus-baseline contrast, which is confounded with natural history and cannot separate the *BR* and *PB* components.
- With no off-drug phase there is no discontinuation-driven vulnerability window; dropout reduces only the precision of the on-drug slope.
- The design is the robustness benchmark but is the wrong design for the predictive-biomarker question.

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593 The open-label design comprises a single path on which every time-
594 point is on-drug. The only within-participant contrast available is
595 the on-drug response against baseline, which is confounded with the
596 natural-history trajectory TV and cannot separate the pharmacologi-
597 cal response BR from the placebo response PB . Because there is no
598 off-drug phase, there is no discontinuation-driven vulnerability win-
599 dow, and informative dropout can do no more than reduce the pre-
600 cision with which the on-drug slope is estimated. The design is con-
601 sequently the most robust to dropout of the four and serves as the
602 robustness benchmark, but its inability to identify the BR -versus- PB
603 split makes it the wrong instrument for the predictive-biomarker ques-
604 tion that motivates the program.

605 [24]

606 **The open-label-plus-blinded-discontinuation design adds a**
607 **silent-placebo phase that identifies the pharmacological-**
608 **versus-placebo split but opens a discontinuation window**
609 **in which worsening-driven dropout removes exactly that**
610 **contrast.**

- 611 • The estimand-defining contrast is the on-drug-versus-post-
612 discontinuation response, which separates BR from PB .
- 613 • Treatment-responders whose symptoms return on the silently
614 switched placebo may drop, mistaking the return for trial
615 failure, which is a worsening-driven dropout located in the
616 discontinuation phase.
- 617 • Because the lost phase carries the identifying contrast, dropout
618 here biases $\hat{\beta}_{bm:D}$ rather than merely inflating its variance.

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621 The open-label-plus-blinded-discontinuation design retains the open-
622 label phase and appends a blinded discontinuation phase in which the
623 active drug is silently replaced by placebo. The discontinuation phase
624 supplies the identifiability that the bare open-label design lacks: the
625 contrast between the on-drug response and the post-discontinuation re-
626 sponse separates the pharmacological component BR from the placebo
627 component PB . That same phase is the design's vulnerability win-
628 dow. A treatment-responder whose symptoms return on the silently
629 switched placebo may interpret the return as trial failure and drop
630 out, a worsening-driven dropout concentrated in the discontinuation
631 phase. The damage is more than a loss of precision: because the
632 removed phase carries the very contrast that defines the estimand,
633 dropout here produces a systematic negative bias in $\hat{\beta}_{bm:D}$, the effect

634 quantified in¹ Section 3.4 and reproduced in Study 2.

635 [25]

636 **The traditional crossover design has two randomization paths**
637 **and two transition windows, and is robust to worsening-**
638 **driven dropout but exposed to improvement-driven dropout**
639 **at each crossover.**

- 640 • The drug-first and placebo-first paths each carry a drug-
641 block-versus-placebo-block contrast that identifies the within-
642 participant treatment effect.
- 643 • A responder who feels well on the drug block and worsens after
644 crossing to placebo may drop, an improvement-driven dropout
645 that removes the second block and one crossover contrast.
- 646 • The two transition windows make the design more exposed than
647 the single-window OL+BDC under improvement-driven dropout,
648 though comparably robust under worsening-driven dropout.

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651 The traditional crossover design has two randomization paths, one
652 beginning on drug and crossing to placebo and the other the reverse,
653 each composed of two blocks separated by a transition. The identifying
654 contrasts are the drug-block-versus-placebo-block differences carried
655 within each path. The design has two transition windows, one per
656 crossover, and its exposure is asymmetric in the dropout mechanism.
657 Under worsening-driven dropout it is comparatively robust, because
658 worsening tends to occur on the placebo block and a dropout there still
659 leaves the drug-block observations that anchor the contrast. Under
660 improvement-driven dropout it is exposed: a responder who feels well
661 on the drug block and then worsens after crossing to placebo may
662 drop, removing the second block and with it one of the two crossover
663 contrasts. The two transition windows therefore make the crossover
664 design more vulnerable than the single-window OL+BDC design when
665 the operative mechanism is improvement-driven.

666 [26]

667 **The hybrid N-of-1 design has the largest number of identify-**
668 **ing contrasts at risk and the most vulnerability windows, but**
669 **recovers power through the happy-accident selection effect**
670 **that preferentially preserves the most informative crossover**
671 **blocks.**

- 672 • An open-label run-in followed by a randomized multi-block
673 crossover gives four randomization paths and the widest set of
674 vulnerability windows.

- Participants with a clear open-label response are resolved early, so those continuing to the later randomized blocks are enriched for inconclusive open-label response.
- Biased dropout therefore preferentially preserves the most informative crossover blocks, partially offsetting the raw power loss from the larger vulnerability surface.

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The hybrid N-of-1 design of¹ combines an open-label run-in, which establishes each participant's apparent response, with a subsequent randomized multi-block crossover. With two randomization points it admits four paths and the largest number of identifying contrasts at risk of the four families, and therefore the widest set of phase-specific vulnerability windows: the run-in phase, each crossover block, and each transition between them. On raw power this makes the hybrid design the most dropout-exposed. The exposure is partly offset, however, by a selection effect that¹ terms a 'happy accident'. Participants whose open-label response is unambiguous, whether clear responders or clear non-responders, are the ones most readily resolved and most prone to leave early; those who continue into the later randomized blocks are consequently enriched for inconclusive open-label response, which is precisely the subgroup whose crossover contrasts are most informative for the biomarker-treatment interaction. Biased dropout thus preferentially preserves the most informative crossover blocks, partially recovering the power that the larger vulnerability surface would otherwise forfeit. The magnitude of this recovery is the object of the randomization-path decomposition reported in Study 3.

5 Simulation design

[27]

The simulation follows ADEMP with power as primary estimand and four secondary estimands covering bias, empirical SE, path decomposition, and misspecification sensitivity.

- Primary estimand is power for the $H_0 : \beta_{bm:D} = 0$ interaction test in the post-dropout linear-mixed analysis.
- Bias is reported in two modes: relative to the calibrated true value and relative to the full-data gold standard.
- Secondary estimands include path-conditional power (Study 3) and dropout-pattern misspecification sensitivity (Study 4).

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The simulation program follows the ADEMP framework of². The primary estimand is the power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ in the linear-mixed analysis fitted to the post-dropout retained data. Secondary estimands are the bias of $\hat{\beta}_{bm:D}$ (reported in two reference modes: relative to the calibrated true value and relative to the gold-standard estimate from the full uncensored data), the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates, the path-conditional power decomposition that quantifies the ‘happy accident’ selection effect (Study 3), and the analysis-method sensitivity to dropout-pattern misspecification (Study 4).

Notation for the moderation parameter. The data-generating process used here is the mean-moderation architecture, so the biomarker-moderation parameter is written β_{bm} , the dimensionless multiplier scaling the treatment effect by the standardized biomarker b_i . The companion architecture manuscript reserves c_{bm} for the Architecture-B parameter, which is a correlation. The reference value 0.45 denotes a matched moderation strength under either architecture, but the symbols are not interchangeable.

[28]

Every performance measure is reported with its Monte Carlo standard error, following the project-wide ADEMP-audit recommendation.

- Power and Type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$.
- Bias and empirical SE of $\hat{\beta}_{bm:D}$ carry the standard MCSE of the within-replicate mean.
- Reporting MCSEs alongside every estimate is a project-wide ADEMP-audit requirement.

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Performance measures are reported with MCSEs alongside every estimate, addressing the project-wide ADEMP-audit recommendation. Power and Type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$; bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard MCSE of the within-replicate mean.

[29]

Four pre-registered studies fix the simulation scope before production runs: power, bias, path decomposition, and misspecification sensitivity.

- Study 1 (180 cells) reproduces and extends Hendrickson Figure 4A for power as a function of design and dropout pattern.

- Study 2 (120 cells) addresses two-mode bias quantification; Study 3 (12 cells) examines the happy-accident path decomposition.
- Study 4 (60 cells) evaluates sensitivity to dropout-pattern misspecification at the analysis stage.

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The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/informative-dropout-by-design/0` which fixes four studies before production runs. Study 1 (180 cells) covers power as a function of design and dropout pattern, reproducing and extending¹ Figure 4A. Study 2 (120 cells) addresses two-mode bias quantification. Study 3 (12 cells with path- conditional decomposition) examines the ‘happy accident’ randomization-path selection effect. Study 4 (60 cells) evaluates sensitivity to dropout-pattern misspecification at the analysis stage.

(Forthcoming detail. Pmsimstats infrastructure with the existing R/`censordata.R` hazard-based dropout module. Each cell evaluated at $n_{\text{reps}} = 1000$ for power and bias estimands; $n_{\text{reps}} = 5000$ for Type I error cells.)

6 Results

[30]

The prototype ran 500 replicates per cell across 16 cells with 100 percent convergence and achieved dropout fractions close to targets.

- 8-core `parallel::mclapply` completed 8,000 fits in 421 s with 100 percent convergence.
- Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime; hybrid cells saturate near 1.000 and OL cells near zero.
- Achieved dropout fractions tracked targets: approximately 0.25 for `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`.

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The simulation program was executed at 500 replicates per cell across the 16-cell design-by-dropout-pattern grid using 8-core `parallel::mclapply` in 421 s. Convergence was 100 percent across all 8,000 fits in every cell. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/09-dropout-summary.txt`. Cell MCSE on power runs 0.014 to 0.019 in the binding 0.85 regime, with the hybrid cells effectively saturated near 1.000 and the OL cells near zero. Achieved dropout fractions tracked targets closely: approx-

imately 0.25 for both MCAR_25 and biased_25, approximately 0.40 for biased_40.

6.1 Power across designs and dropout patterns

[31]

Power results reproduce the Hendrickson happy-accident ordering: hybrid dominates, OL falls to nominal alpha, with intermediate designs losing power monotonically under heavier dropout.

- Hybrid (approximately 1.00) > OL+BDC (approximately 0.95) > traditional crossover (approximately 0.85) > OL (nominal alpha 0.02–0.05).
- Hybrid dominates because post-week-8 branching produces within-subject and across-path contrasts that survive even severe informative dropout.
- The ordering reproduces the Hendrickson happy-accident randomization-path argument at the wider 16-cell grid.

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We begin with the power results. The four designs reproduce the¹ happy-accident randomization-path ordering at the wider 16-cell grid: the hybrid design (power approximately 1.00) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal Type I level near 0.02 to 0.05). The hybrid four-path design dominates because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc} that survive even severe informative dropout. Cell-level power values, percent $\pm$ MCSE:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	2.2 $\pm$ 0.7	4.6 $\pm$ 0.9	3.4 $\pm$ 0.8	4.4 $\pm$ 0.9
OL+BDC	95.0 $\pm$ 1.0	91.2 $\pm$ 1.3	92.4 $\pm$ 1.2	89.4 $\pm$ 1.4
traditional crossover	85.4 $\pm$ 1.6	81.0 $\pm$ 1.8	82.4 $\pm$ 1.7	76.8 $\pm$ 1.9
hybrid	99.8 $\pm$ 0.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	99.4 $\pm$ 0.3

[32]

The OL row at nominal alpha is a structural property of the mean-moderation architecture, not a dropout artifact, and functions as an implicit null check.

- Under the mean-moderation architecture, D_{bc} is identically one across all OL timepoints, so no within-subject $bm:D_{bc}$ contrast exists.

- With no interaction information, the fitter’s rejection rates collapse to the nominal Type I rate across all dropout cells.
- The OL row is therefore an implicit structural null check, not a power estimate, and should not be interpreted as dropout robustness.

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The OL row sits at nominal alpha under every dropout pattern because the mean-moderation architecture’s level shift on BR at on-drug time-points is uniform across OL1–OL8: with D_{bc} identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal Type I rate. This is a structural property of the OL design under the mean-moderation architecture, not a dropout artifact; the OL row therefore functions as an implicit null check rather than a power estimate.

[33]

Dropout resilience and absolute power coincide in ranking: hybrid is most robust and most powerful, traditional crossover is least robust and least powerful of the three powered designs.

- Absolute power loss from no-dropout to `biased_40`: hybrid 0.4 pp, OL+BDC 5.6 pp, traditional crossover 8.6 pp.
- The resilience ordering (hybrid > OL+BDC > crossover) coincides with the absolute-power ordering.
- The most powerful design is also the most dropout-robust, a non-trivial alignment with practical design-selection implications.

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Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the `biased_40` cell, runs: hybrid (loss approximately 0.4 percentage points) above OL+BDC (loss approximately 5.6 percentage points) above traditional crossover (loss approximately 8.6 percentage points). Of the three powered designs, the hybrid is most robust and the traditional crossover most vulnerable; it is worth noting that the resilience ordering coincides with the ordering of absolute power, the most powerful design also being the most robust.

6.2 Bias of the biomarker-treatment interaction estimate

[34]

The mean estimated interaction coefficient is stable across dropout cells within each design, indicating no systematic

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bias in the estimand.

- $\hat{\beta}_{bm:D_{bc}}$ shows no systematic trend across none, MCAR_25, biased_25, and biased_40 cells within any design.
- Stability holds for the three powered designs (OL+BDC, crossover, hybrid); the OL design is uninformative on bias.
- The absence of bias in the estimand implies the power degradation traces to standard-error inflation, not bias.

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We turn next to the bias of the biomarker-treatment interaction estimate. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells within each design, indicating no systematic bias in the estimand under informative dropout in this regime:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	-\$0.09	-\$0.10	-\$0.09	-\$0.10
OL+BDC	-\$2.48	-\$2.48	-\$2.47	-\$2.45
traditional crossover	-\$2.24	-\$2.28	-\$2.25	-\$2.21
hybrid	-\$2.46	-\$2.46	-\$2.46	-\$2.45

[35]

Power degradation under informative dropout traces to standard-error inflation, not bias in the estimand: empirical SEs expand modestly under biased_40 dropout.

- The largest within-design bias shift (approximately 0.04 for traditional crossover) is within cell-level MCSE.
- Empirical SEs expand under biased_40: OL+BDC 0.628 to 0.660, crossover 0.766 to 0.779, hybrid 0.411 to 0.454.
- The power loss in Section 5.1 is therefore attributable to SE inflation rather than systematic bias.

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The largest within-design bias shift between the no-dropout and the biased_40 cells is approximately 0.04 (traditional crossover), well within the cell-level MCSE on the mean (≤ 0.04 for all designs). The power degradation visible in Section 5.1 therefore traces to inflation of the standard error of $\hat{\beta}_{bm:D_{bc}}$, not to systematic bias in the estimand. Empirical SEs (last column of 09-dropout-summary.txt) expand modestly under biased_40 dropout: from 0.628 to 0.660 for OL+BDC; from 0.766 to 0.779 for traditional crossover; and from 0.411 to 0.454 for hybrid.

6.3 Randomization-path decomposition

[36]

Path-stratified decomposition of the happy-accident effect is scoped for a production rerun; the current prototype provides only marginal results consistent with the hypothesis.

- The current prototype aggregates across paths within each design and cannot attribute robustness to specific surviving paths.
- The Hendrickson happy-accident hypothesis is consistent with the marginal results but requires path-tracked data for a formal test.
- The follow-up extension requires recording `path_id` per replicate and adding path-stratified $\hat{\beta}_{bm:D_{bc}}$ summary logic.

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We note that decomposition of the headline power loss into randomization-path-specific contributions is scoped for a production-grade rerun with per-path-id tracking added to the driver. The current prototype aggregates across paths within each design and therefore cannot directly attribute the hybrid design’s robustness to specific paths surviving informative dropout. The substantive¹ ‘happy accident’ hypothesis (that informative dropout preferentially preserves the most-informative crossover blocks) is consistent with the present marginal results, but a formal test against the alternative (dropout uniform across paths) requires the path-tracked data that the current prototype does not produce. The follow-up driver extension is relatively straightforward at the simulation level, requiring only that `path_id` be recorded per replicate; the bottleneck is in the analysis layer, where path-stratified $\hat{\beta}_{bm:D_{bc}}$ distributions need to be summarized against the cell-pooled estimate.

6.4 Sensitivity to dropout-pattern misspecification

[37]

The `biased_25` versus `MCAR_25` contrast isolates the bias mechanism from the rate effect; within-design differences are small and statistically indistinguishable at 500 replicates per cell.

- OL+BDC: `biased_25` minus `MCAR_25` equals +1.2 pp (CI $[-2.4, +4.8]$); crossover: +1.4 pp (CI $[-3.5, +6.3]$).
- Hybrid is saturated; OL is at null; all differences lie within or near 1.96 cell-MCSE.
- Resolving the distinguishability question would require approximately 2,000 replicates per cell at the current MCSE.

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945 The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout
946 isolates the bias mechanism from the rate effect. The within-design
947 differences are small in absolute terms: OL+BDC `biased_25` power
948 minus `MCAR_25` power equals +1.2 percentage points (CI $[-2.4, +4.8]$);
949 traditional crossover equals +1.4 percentage points (CI $[-3.5, +6.3]$);
950 hybrid is approximately zero (saturated; bracketed); and OL equals
951 -1.2 percentage points (null cells; bracketed).

952 [38]

953 **The dominant power determinant is dropout rate rather than**
954 **bias mechanism; mechanism effects are not statistically dis-**
955 **tinguishable at 500 replicates per cell.**

- 956 • Moving from 25% to 40% biased dropout costs 3.0 pp (OL+BDC),
957 5.6 pp (crossover), 0.6 pp (hybrid).
- 958 • The crossover rate-effect loss is clearly detectable above MCSE;
959 the OL+BDC loss is borderline.
- 960 • Resolving the `biased_25` versus `MCAR_25` mechanism question
961 requires approximately 2,000 replicates per cell, deferred to a
962 production rerun.

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965 All four within-design differences lie within 1.96 cell-MCSE and
966 are not statistically distinguishable from zero at this sample size.
967 The dominant effect on power is the dropout rate rather than the
968 bias mechanism: moving from 25 percent to 40 percent biased
969 dropout costs 3.0 percentage points (OL+BDC), 5.6 percentage
970 points (traditional crossover), and 0.6 percentage points (hybrid),
971 with the crossover loss clearly detectable above MCSE. We note that
972 resolving the `biased_25` versus `MCAR_25` distinguishability question
973 with statistical clarity would require approximately 2,000 replicates
974 per cell at the current MCSE, which we leave to a production rerun.

975 7 Design-by-dropout recommendation

976 [39]

977 **Three classes of anticipated dropout mechanism are mapped**
978 **to preferred N-of-1 design choices based on the Section 5**
979 **simulation results.**

- 980 • The mapping is indexed by anticipated overall dropout rate
981 within each mechanism class.

- The recommendation draws directly on the power and bias results from the 16-cell prototype simulation.
- The table is intended as practical guidance for trial designers selecting among the four canonical N-of-1 families.

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In this section we shall map three classes of anticipated dropout mechanism (no informative dropout, informative-by-rate, and informative-by-mechanism) to preferred N-of-1 design choices, drawing on the Section 5 results.

Anticipated dropout	Preferred design	Rationale
None or low MCAR ($d \leq 10\%$)	hybrid	Saturated power; no penalty for hybrid's complexity
Moderate MCAR ($d \approx 25\%$)	hybrid	Hybrid retains saturated power; OL+BDC loses 4 pp; crossover loses 4 pp
Heavy MCAR or biased ($d \geq 40\%$)	hybrid	Hybrid at 99% power; OL+BDC drops to 89%; crossover drops to 77%
Worsening-driven (any rate)	hybrid or OL+BDC	OL+BDC's blinded discontinuation phase carries the bias signal explicitly
Improvement-driven (any rate)	OL+BDC	Crossover loses signal when responders drop early; OL+BDC absorbs
Information-poor (OL only)	not recommended	Architecture-A OL has no within-subject $bm:D_{bc}$ contrast

[40]

The dominant recommendation message is that the hybrid design's power advantage extends to dropout robustness across all three mechanism classes.

- Hybrid is preferred under all dropout rate and mechanism classes except improvement-driven, where OL+BDC is preferred.
- Traditional crossover loses approximately 9 percentage points under `biased_40` and is the least dropout-robust powered design.
- The OL design lacks the within-subject D_{bc} contrast under the mean-moderation architecture and does not warrant inclusion in the primary recommendation.

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The dominant message from the table is that the hybrid Hendrickson design's power advantage under no dropout extends to robustness against informative dropout. The traditional crossover, despite its strong no- dropout power, loses approximately 9 percentage points

under `biased_40` dropout and is the least dropout-robust of the three powered designs. The OL design lacks the within-subject D_{bc} contrast under the mean-moderation architecture and serves only as a null check at the present prototype’s resolution; it therefore does not warrant inclusion in the primary recommendation.

8 Discussion

8.1 Prototype Monte Carlo confirmation

[41]

The 500-replicate prototype confirms 100 percent convergence across all 8,000 fits with achieved dropout fractions close to targets.

- Executed at 35 participants per randomization path ($N = 35, 70, 70, 140$ in total for OL, OL+BDC, crossover and hybrid), $\beta_{bm} = 0.45$, the mean-moderation architecture on 8 cores in 421 s with 100 percent convergence.
- Achieved dropout: approximately 0.25 for `MCAR_25` and `biased_25`; approximately 0.40 for `biased_40`.
- Cell MCSE on power runs 0.014–0.019; hybrid saturates near 1.000.

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A 500-replicate-per-cell prototype run on the full 16-cell design-by-dropout grid (design $\in \{\text{OL, OL+BDC, traditional crossover, hybrid}\} \times \text{dropout pattern} \in \{\text{none, MCAR 25\%, biased 25\%, biased 40\%}\}$) was executed at $\beta_{bm} = 0.45$ under the mean-moderation architecture using 8-core `parallel::mclapply`. Wall clock was 421 s; convergence was 100 percent across all 8,000 fits.

The prototype driver fixed the sample size at 35 participants *per randomization path* rather than in total. Because the four designs have different path counts (one for OL, two for OL+BDC and traditional crossover, four for hybrid), the totals in this run are $N = 35, 70, 70$ and 140 respectively. The compendium convention is that N is the total, so these are the figures to quote; the design comparison below is not matched on N , and the implications of that are taken up in the next subsection. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP summary is at `09-dropout-summary.txt`. Achieved dropout fractions tracked targets closely: approximately 0.25 for both `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`. Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime, with

the hybrid cells effectively saturated near 1.000.

[42]

The prototype reproduces the Hendrickson happy-accident power ordering: hybrid > OL+BDC > crossover > OL at nominal alpha.

- Power ordering: hybrid approximately 1.00, OL+BDC approximately 0.95, crossover approximately 0.85, OL at nominal 0.02–0.05.
- Hybrid dominates because post-week-8 branching produces within-subject and across-path D_{bc} contrasts that survive severe informative dropout.
- The ordering holds across all four dropout-pattern cells, confirming the Hendrickson happy-accident argument at wider parameter ranges.

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The four designs reproduce the¹ happy-accident randomization-path ordering at the wider grid: hybrid (approximately 1.00 power) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal Type I level near 0.02–0.05). The hybrid four-path design dominates because its branching after week 8 produces both within- subject and across-path contrasts in D_{bc} that survive even severe informative dropout.

This ordering must be read against the sample sizes it was produced at. Because the driver fixed the per-path count, the hybrid cells carry $N = 140$ against $N = 70$ for OL+BDC and crossover, so the hybrid advantage is confounded with a doubled total and cannot be attributed to the branching structure on this evidence alone. Two comparisons are unaffected. OL+BDC against traditional crossover is matched at $N = 70$, so the 0.95-versus-0.85 gap is a design effect. The OL row is a structural null, as the following subsection shows, and its position does not depend on sample size. Separating the hybrid design’s contribution from its larger sample requires a re-run at matched totals, which has not been done.

[43]

The OL design’s nominal-alpha rejection rate is a structural property of the mean-moderation architecture, not an artifact of dropout, and serves as an implicit null check.

- Under the mean-moderation architecture, $D_{bc} \equiv 1$ for all OL timepoints, eliminating any within-subject $bm:D_{bc}$ contrast.

- The fitter has no interaction information, so rejection rates collapse to nominal alpha across all dropout conditions.
- This confirms the OL row is a structural null check rather than evidence of dropout robustness.

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The OL row sits at nominal alpha under every dropout pattern because the mean-moderation architecture's level shift on BR at on-drug time-points is uniform across OL1–OL8: with D_{bc} identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal Type I rate. This is a structural property of the OL design under the mean-moderation architecture, not a dropout artifact; the OL row therefore functions as an implicit null check, not a power estimate.

[44]

Dropout resilience is dominated by the rate effect, not the bias mechanism; the linear-mixed model with corCAR1 is relatively unbiased under informative dropout in this regime.

- Absolute power loss from no-dropout to `biased_40`: hybrid 0.004, OL+BDC 0.056, crossover 0.086.
- The `biased_25` vs. `MCAR_25` mechanism contrast is within or barely outside 1.96 MCSE; the rate increase from 25 to 40 percent is the dominant effect.
- $\hat{\beta}_{bm:D_{bc}}$ is stable (−2.45 to −2.48), confirming power degradation traces to SE inflation rather than bias in the estimand.

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Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the `biased_40` cell, runs as follows: hybrid (loss approximately 0.004) above OL+BDC (loss approximately 0.056) above traditional crossover (loss approximately 0.086). The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout is small (+0.012, +0.014, −0.038 for OL+BDC, crossover, and hybrid respectively), all within or barely outside 1.96 MCSE. The dominant effect is dropout rate, not bias mechanism: moving from 25 percent to 40 percent biased dropout costs 0.006–0.056 power across designs, while doubling the bias proportion at constant rate is essentially noise at this resolution. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells (−2.45 to −2.48 for OL+BDC and hybrid), suggesting that the linear-mixed model with `corCAR1` is relatively unbiased under informative dropout in this regime; the power degradation traces to standard-error inflation rather than

systematic bias in the estimand.

[45]

Two ADEMP caveats limit the prototype's scope: no formal null cells for non-OL designs, and single-architecture coverage.

- No $\beta_{bm} = 0$ cells were run for OL+BDC, crossover, or hybrid; production reruns should add explicit null cells for each.
- The prototype is single-architecture (the mean-moderation architecture); under the covariance-moderation architecture the OL row carries information via the covariance structure.
- Resolving the `biased_25` vs. `MCAR_25` distinguishability question requires approximately 2,000 replicates per cell.

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Two ADEMP-discipline caveats warrant mention. First, no formal Type I error cells under $\beta_{bm} = 0$ were run for the non-OL designs; the OL row serves as a structural null check by argument, not as a designed null cell. A canonical production run should add explicit $\beta_{bm} = 0$ cells for OL+BDC, crossover, and hybrid. Second, the prototype is single-architecture (the mean-moderation architecture) by specification; under the covariance-moderation architecture (multivariate normal, MVN) the OL row would carry information because the biomarker-treatment signal lives in the covariance structure rather than in a within-subject mean contrast. Resolving the `biased_25` versus `MCAR_25` distinguishability question would, as noted above, require approximately 2,000 replicates per cell at the current MCSE.

9 Conclusions

[46]

The conclusions distil three findings: design-by-dropout differentiation, rate dominance over mechanism, and SE inflation as the power-loss mechanism.

- The four N-of-1 design families differ materially in dropout robustness in ways the published literature has not characterized.
- Dropout rate dominates mechanism as the determinant of power loss; mechanism becomes load-bearing only when rates are comparable.
- Estimand stability under informative dropout is reassuring but also a warning: coverage degrades as SEs widen without the model acknowledging it.

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In conclusion, three points merit emphasis.

[47]

First conclusion: the hybrid design is the most dropout-robust of the four, losing less than half a percentage point under 40 percent biased dropout at the reference parameters.

- At 35 per path ($N = 140$ hybrid, 70 OL+BDC and crossover), $\beta_{bm} = 0.45$, the mean-moderation architecture: hybrid loses < 0.5 pp, OL+BDC loses 5.6 pp, crossover loses 8.6 pp under `biased_40`.
- OL lacks the within-subject D_{bc} contrast under the mean-moderation architecture and is not viable for the biomarker-interaction question.
- The hybrid's robustness extends the Hendrickson happy-accident argument without requiring path-stratified analysis.

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First, the four candidate aggregated N-of-1 trial designs differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterized. At the prazosin/PTSD reference parameters with 35 participants per randomization path ($N = 140$ in total for the four-path hybrid and $N = 70$ for the two-path designs), $\beta_{bm} = 0.45$, and the mean-moderation architecture, the hybrid Hendrickson design is the most dropout-robust of the four, losing less than half a percentage point of power under 40 percent biased dropout. The OL+BDC design loses approximately 5.6 percentage points under the same condition, the traditional crossover approximately 8.6 percentage points, and the OL design lacks the within-subject D_{bc} contrast required for the biomarker-treatment interaction under the mean-moderation architecture. The hybrid's robustness extends and corroborates the¹ happy-accident randomization-path argument without requiring path-stratified analysis to support the substantive claim.

[48]

Second conclusion: dropout rate is the dominant power determinant; mechanism is load-bearing only when rates are comparable across competing design candidates.

- The `biased_25` vs. `MCAR_25` contrast at fixed 25 percent dropout is below MCSE detection at 500 replicates.

- Moving to 40 percent biased dropout produces clear losses in OL+BDC and crossover; hybrid is unaffected.
- Trial designers should evaluate designs against anticipated dropout rate first, and mechanism only secondarily.

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Second, the dominant determinant of the design-by-dropout interaction, in this prototype's evaluation, is the dropout rate rather than the dropout mechanism. The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout produces within-design differences below MCSE detection at 500 replicates per cell; moving to 40 percent biased dropout, by contrast, produces clear power losses in every design except the hybrid. Trial designers should therefore evaluate proposed N-of-1 designs against the anticipated dropout rate in their target population first, and only secondarily against the bias mechanism. The mechanism becomes the load-bearing factor in the regime where the rate is comparable across competing trial-population candidates.

[49]

Third conclusion: estimand stability is reassuring but also a coverage warning, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

- The largest within-design shift in $\hat{\beta}_{bm:D_{bc}}$ is below 0.04 across the no-dropout to `biased_40` range.
- Power degradation traces to SE inflation; the inferential target is preserved under informative dropout in this regime.
- Nominal CI coverage may understate true uncertainty: the coefficient distribution widens but the model does not acknowledge it.

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Third, the mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells, with the largest within-design shift below 0.04 across the no-dropout to `biased_40` range. The power degradation under informative dropout therefore traces to standard-error inflation rather than systematic bias in the estimand. This finding is reassuring for the validity of the aggregated N-of-1 framework in the presence of informative dropout, in the sense that the inferential target is preserved, but it is also a warning to trialists that nominal coverage of confidence intervals may understate the true uncertainty in the dropout regime, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

[50]

Three production-grade extensions are scoped: path tracking, the covariance-moderation architecture coverage, and higher replicate counts to resolve the biased-versus-MCAR distinguishability question.

- Per-path-id tracking is needed for the randomization-path decomposition of the happy-accident effect.
- The covariance-moderation architecture coverage is needed to interpret the OL row as a power estimate rather than a structural null check.
- Resolving the `biased_25` vs. `MCAR_25` question requires 2,000 or more replicates per cell.

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Production-grade extensions of the prototype (per-path tracking for the randomization-path decomposition, the covariance-moderation architecture for the OL design’s null-check elevation, and 2,000 or more replicates per cell to resolve the biased-versus-MCAR distinguishability question) are scoped in Section 5 of this manuscript and represent the natural follow-up to the present proof-of-concept evidence.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project’s `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the

original-extended simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The informative-dropout module is at `R/censordata.R`.

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/09-dropout-replicates.rds`. Morris ADEMP cell-level summaries are at the companion `09-dropout-summary.txt`.

Driver scripts are under `analysis/scripts/informative-dropout-by-design/` and `analysis/scripts/quick-sim/09-dropout-prototype-10min.R`. The pre-registered ADEMP plan is at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-registra`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/09-informative-dropout-by-design/`.

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